Review Pulse

We trained two sentiment classifiers on 8,000 Amazon reviews and built a live web app to compare them.

INTELLIGENT SYSTEMS (ISY503)
ASSESSMENT 3

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May, 2026

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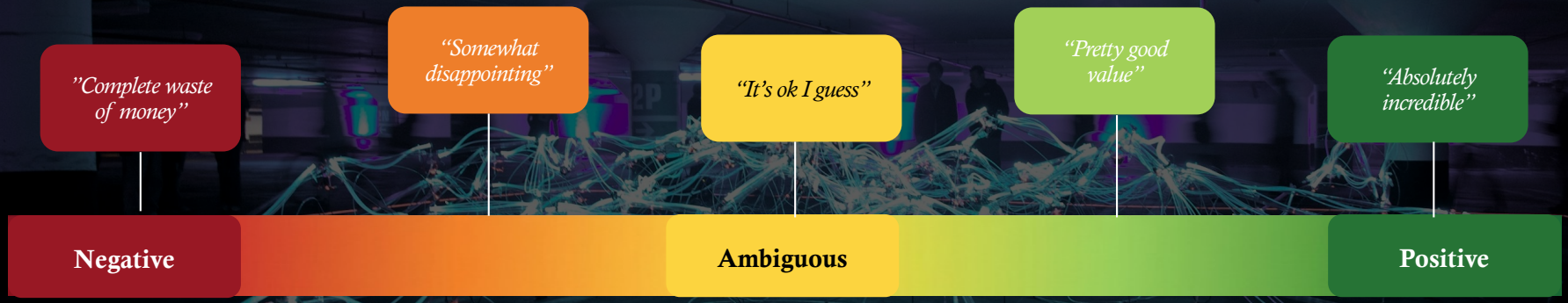
PROBLEM STATEMENT

The Sentiment Spectrum – what our model must learn to classify



- Sentiment analysis: classifying text as positive or negative opinion
- Real-world value: product feedback loops, brand monitoring, recommendation systems
- Challenge: reviews vary in **length** (2–800 words), **domain**, **writing style**, **negation**, and **sarcasm**
- Goal: build, compare, and deploy two approaches — classical ML vs neural

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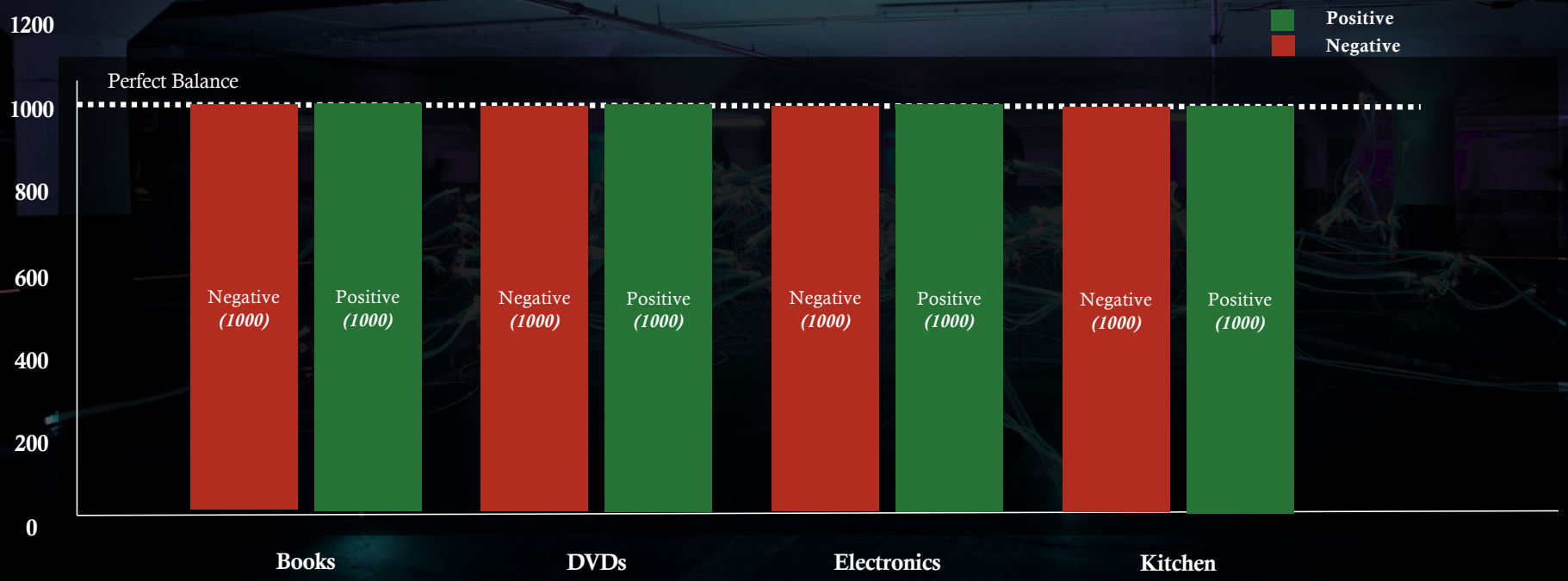
DATASET

Balance – 1,000 reviews per domain per class



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PREPROCESSING PIPELINE

Raw .review Files – pseudo-XML format

↓

BeautifulSoup Parser – extract review_text + rating

↓

Label Audit – flag 3-star and rating/label conflicts

↓

clean_text – lowercase, HTML strip, negation expansion, remove punctuation

Wasn't good → was not good

↓

remove_outliers – drop < 10 or > 500 words (279 removed)

↓

Stratified Split – 70% train / 15% val / 15% test (seed = 42)

↓

train = 5,404 / val = 1,158 / test = 1,159

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ARCHITECTURE OVERVIEW

Challenge	*Example*	*Mitigation*
Security & Privacy	PII + payment data	IAM least-privilege, MFA, encryption, shared responsibility model
Cost Volatility	Pay-as-you-go can spiral without guardrails	FinOps: budget alerts, resource tagging, rightsizing, reserved pricing
Vendor Lock-in + Skills Gap	Migration cost + mindset shift	Portability-first (containers, standard DBs), targeted cloud upskilling

TRAINING

Cloud Service Models

IaaS
Compute + Config
Flexibility

PaaS
Managed DB, load balancing,
serverless

SaaS
Limited customization

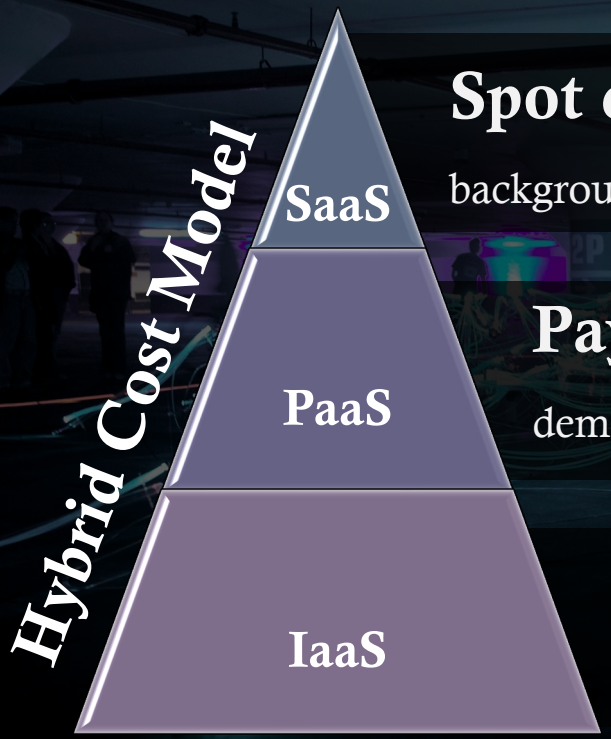
Cloud Deployment Models

Private cloud
High cost, limited elasticity

Hybrid cloud
Premature complexity

Public cloud
Scalable, automated,
cost-effective

RESULTS



Spot capacity

background jobs + analytics pipelines

Pay-as-you-go

demand spikes via Auto Scaling

Reserved

stable tiers (web/app servers, databases)

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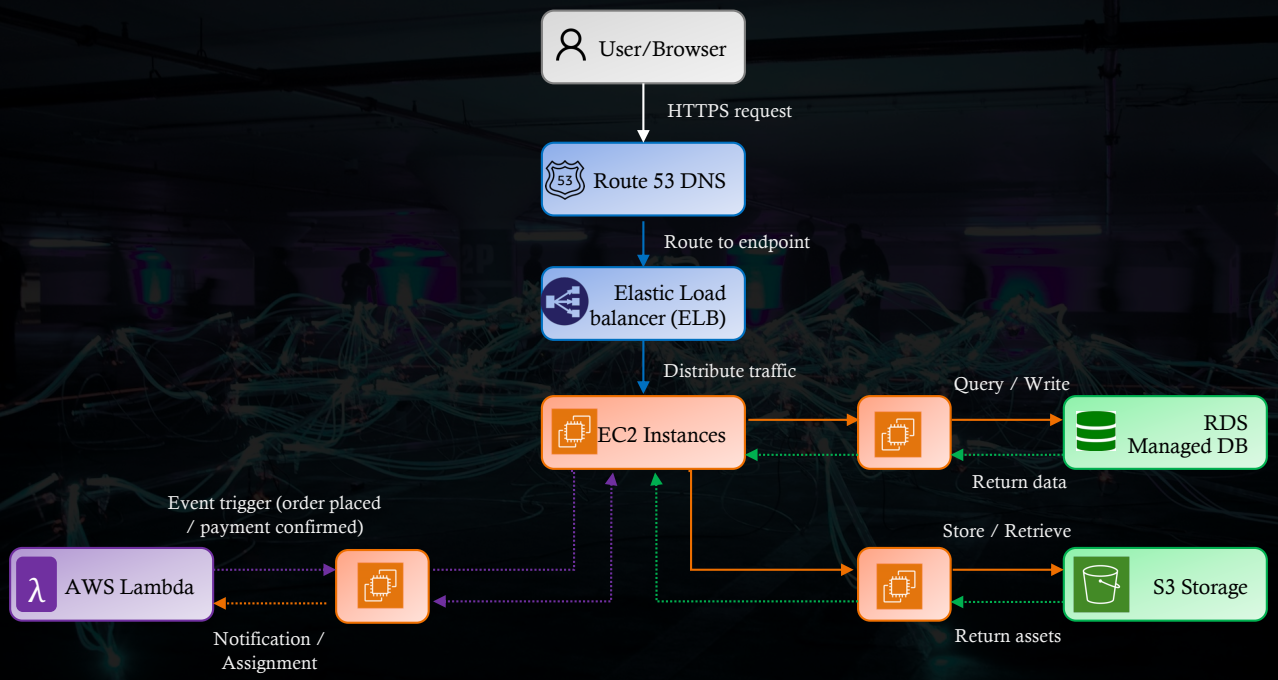
Error Analysis

Provider	Ecosystem Fit	Load Balancing	Serverless	ABC Alignment
aws	Broadest managed-service catalogue	ELB – native Route53 integration	Lambda – event-driven, zero idle cost	Best fit – Route 53 already in stack
Azure	Microsoft / enterprise-aligned	Application gateway – extra config needed	Azure Functions – strong but separate ecosystem	No existing Azure signals in ABC
Google Cloud	Analytics and ML-first	Cloud load balancing – GKE-oriented	Cloud Run / Functions – container first	No analytics-heavy workloads in ABC now

Live Demo



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ETHICS & LIMITATIONS

Public cloud | IaaS+PaaS | Hybrid cost model | AWS

Key Outcomes Unlocked:

- 80% cost reduction (measured service)
- 10x surge absorbed (rapid elasticity)
- Zero-headcount scaling (resource pooling + Auto Scaling)
- Workflow automation (Lambda event triggers)

FUTURE WORK

Public cloud | IaaS+PaaS | Hybrid cost model | AWS

Key Outcomes Unlocked:

- 80% cost reduction (measured service)
- 10x surge absorbed (rapid elasticity)
- Zero-headcount scaling (resource pooling + Auto Scaling)
- Workflow automation (Lambda event triggers)

SUMMARY / QUESTIONS

Public cloud | IaaS+PaaS | Hybrid cost model | AWS

Key Outcomes Unlocked:

- 80% cost reduction (measured service)
- 10x surge absorbed (rapid elasticity)
- Zero-headcount scaling (resource pooling + Auto Scaling)
- Workflow automation (Lambda event triggers)

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Thank you!

INTELLIGENT SYSTEMS (ISY503)
ASSESSMENT 1

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